

FasalDisha —

AI Crop Price Forecasting & Market Routing System (Round 2 v2.0)
Official Judge Demonstration Flow & Evaluation Playbook

1. Executive Value Proposition & Opening Pitch

The Core Question: 'Given my commodity, quantity, coordinates, price forecast, transport cost, and active risks — where should I sell, and should I sell now or wait?'

FasalDisha is not a passive price board. It is an end-to-end **forecasting + market-routing + risk-aware decision engine** that empowers farmers with transparent, quantity-aware, and risk-adjusted market options.

2. Quick Server Startup (Run in 2 Terminals)

Component	Port / URL	Startup Command (PowerShell)
Backend (FastAPI)	http://127.0.0.1:8000	\$env:PYTHONPATH='backend;.' ; uvicorn app.main:app --port 8000
Frontend (React+Vite)	http://localhost:5173	cd frontend ; npm run dev

3. Step-by-Step 3-Minute Judge Demonstration Flow

Step A: Hero Scenario — Pune Onion (Deterministic Seeded Weather Risk Override)

- Action:** On the form, click preset button '**Scenario A: Pune Onion (Risk Override)**' and click '**Run Analysis**'.
- Context:** Maharashtra / Pune, Coordinates (18.52° N, 73.85° E), Onion (25 Quintals), Radius 120 km.
- Key Highlight 1 (Recommendation Banner):** Show that the base forecast decision was **HOLD** (prices rising), but the final recommendation is **SELL EARLY DUE TO RISK** due to active severe waterlogging.
- Key Highlight 2 (Honest Weather Banner):** Show the red alert banner labeled '**Source: SEEDED**' (clearly disclosing demo scenario).
- Key Highlight 3 (ML Forecast & Peak Alert):** Show 7-day horizons (Day 1: ₹2,390, Day 3: ₹2,480, Peak Day 6: ₹2,650) with active **PEAK ALERT (+12.8%)** and historical series provenance (45 days).
- Key Highlight 4 (Dynamic Mandi Rankings):** Show 7 mandis discovered dynamically with #1 APMC Chakan (Risk-Adjusted Return: ₹60,428 after ₹1,643 transport).
- Key Highlight 5 (Synthetic Buyer Intelligence):** Expand card to show 4 active buyers (Demand: MEDIUM, Offer: 77%, Reliability: 82.5%) labeled '**SYNTHETIC DEMO DATASET**'.
- Key Highlight 6 (Explainability Breakdown):** Show 70% Risk-Adj Return + 20% Buyer Signal + 10% Data Quality formula.

Step B & C: Group-Wise Perishability Comparison (Tomato vs Wheat)

- Action:** Click 'Modify Parameters', then select **Scenario B: Tomato** (Highly Perishable) and run.
- Highlight:** Tomato has high spoilage risk, enforcing an immediate **SELL_NOW** recommendation.
- Action:** Return and select **Scenario C: Wheat** (Non-Perishable / Durable Cereal).
- Highlight:** Durable grains under low operational risk permit holding to capture expected peak prices.

Step D: Cross-Boundary Market Routing (Ahmedabad Cotton)

- Action:** Select **Scenario D: Ahmedabad Cotton** (150 km search radius).
- Highlight:** Point to '**Cross-Boundary Enabled**' badge. Show how coordinate-based radius discovery finds mandis across district/state lines, eliminating artificial administrative barriers.

4. Instant Judge Q&A; Cheat-Sheet (SSOT Ground Truth)

Judge Question	Winning Answer & SSOT Architecture Reference
Why is this mandi ranked #1?	Calculated transparently via <code>ranking_service.py</code> : 70% Risk-Adjusted Net Return (expected revenue minus distance transit cost and risk penalty) + 20% Synthetic Buyer Signal + 10% Data Quality .
Are the buyers real? Where do they come from?	No, they are explicitly labeled SYNTHETIC. We aggregate active buyer records (demand level, offer strength, reliability) from our catalog to model market liquidity without falsely claiming a live marketplace.
How does weather change the decision?	Weather feeds a separate risk layer (<code>risk_service.py</code>). When severe meteorological or transit risk is active, the engine overrides a HOLD decision to SELL_EARLY_DUE_TO_RISK to prevent spoilage.
Why not limit search to the farmer's district?	District is just an entry context. A farmer near a border might have a much higher-paying mandi 20 km away in a neighboring district. Routing is strictly coordinate and radius-driven .
Is the forecast model live or precomputed?	Our system supports pooled ML models (<code>forecast_engine.py</code>) with contract-compliant precomputed fallbacks. We honestly disclose <code>modelType: PRECOMPUTED</code> and <code>historyClassification: SEEDED</code> on screen.

5. Judge-Proof Acceptance & Verification Matrix

Proof Requirement	UI Element / Proof Anchor	Status
Weather Impact	WeatherAlert banner + Seeded Source label + Risk Override	PASS
Group Perishability	3-level perishability badge + storage urgency logic	PASS
Buyer Intelligence	Active buyer count + SYNTHETIC label + demand metrics	PASS
Best Location Selection	Dynamic ranked list + 70/20/10 formula explainability	PASS
ML Forecast & Peak Alert	7-day horizons + PEAK ALERT banner + 45d history basis	PASS
Decision Audit Trail	Judge-Proof Audit Panel with 4 transparent proof cards	PASS